Jahy Luu

832-917-8489 | jahyluu.swe@gmail.com | jahyluu.dev | linkedin.com/in/jahy-luu | github.com/jahyluu

Education

University of Houston **May 2026**
Computer Science B.S. *Houston, TX*

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, C++, SQL, GLSL, HTML/CSS
Frameworks & Libraries: Spring Boot, React, Node.js, Three.js, WebGL, Tailwind CSS, Selenium
AI/ML: LangChain, Llama, Claude API, RAG (ChromaDB), Scikit-learn, Pandas, NumPy
Cloud & Data: AWS (EC2, S3, Lambda, RDS, SageMaker), PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ, Firebase
DevOps & Testing: Git, Docker, GitHub Actions (CI/CD), Gradle, JUnit, JaCoCo, Postman

Work Experience

Software Engineer | *Python, React, Three.js, WebGL, MongoDB, AWS SageMaker* **July 2025 – Present**
Smart Kids University *Houston, TX*

- Engineered a scalable cloud-based robotics simulation engine in collaboration with an overseas university, utilizing **Three.js** and **WebGL shaders**, rendering interactive 3D environments and supporting **450+ concurrent students** during instructor-led virtual lessons
- **Redesigned MongoDB student progress schema** to track performance and error rates across 17 modules, powering a **Llama AI tutor** deployed on **AWS SageMaker** that pinpoints each student's weak points while keeping data in-house for **HIPAA and data-privacy compliance**
- Ported the platform to iPad, **rebuilding the input system for touch controls**, enabling rollout to **3 hospital programs** and **950+ active students**

Software Engineering Intern | *Python, AWS (S3, Lambda), MySQL* **March 2024 – November 2024**
Allure Skin and Wellness *Houston, TX*

- Digitized manual inventory tracking with a c**ustom Python application** and barcode scanner integration, **cutting weekly audit cycles by 15%** and replacing handwritten counts with scan-based entries
- Migrated patient intake and records to a **HIPAA-conscious AWS pipeline (S3, Lambda)** with encrypted storage and role-based access controls, replacing paper workflows and cutting check-in time by 40%

Computer Science Tutor | *Python, Java, C++* **August 2023 – May 2026**
University of Houston *Houston, TX*

- Mentored **100+ CS students** in Data Structures & Algorithms on recursion, dynamic programming, and complexity analysis to boost performance on exams and coding interviews
- Spearheaded **coding workshops** and exam-review sessions with 30+ tutors, raising exam averages by **14 points**

Contract Software Engineer | *Python, AWS EC2* **March 2023 – May 2024**
Fiverr *Remote*

- Built **Python automation pipelines** for 10+ e-commerce clients using **Selenium** to handle real-time stock monitoring, automatic reordering and checkouts, deployed as Dockerized services on **AWS EC2**, saving 10+ hours/week

Projects

Scalable Minecraft Server (Project Hydrius) | *Java, Spring Boot, RabbitMQ, Redis, REST APIs, Bukkit/Spigot*

- Co-developed scalable multiplayer infrastructure with 1 other developer using **Spring Boot**, **REST APIs**, and **RabbitMQ**, enabling real-time event syncing across distributed Minecraft servers with **99.9% uptime**
- Built a JUnit test suite achieving **90% line coverage** (JaCoCo), validating RabbitMQ event handling, Redis cache consistency, and failure-path error handling
- Developed **4 custom Minecraft plugins** backing player economies, live leaderboards, and competitive events for **200+ concurrent players**, caching gameplay states in **Redis** to eliminate per-action database reads

3D Interactive Portfolio | *TypeScript, React, Three.js, React Three Fiber, WebGL, GLSL, Vite, Vercel*

- Built a 3D web app in **React Three Fiber**, **Three.js** and **TypeScript**, cutting **WebGL** draw calls **58%** (2,651 to 1,106 per frame) via mesh merging, shadow-caster culling and throttled shadow maps
- Cut initial JavaScript **86%** (627 → 88 KB gzipped) via **Vite** code splitting and lazy loading; precompiled 37 **GLSL** shaders behind an **OffscreenCanvas Web Worker** intro; deployed via **Vercel CI/CD**

Machine Learning Prediction Model | *Python, Scikit-learn, Pandas, NumPy*

- Trained gradient-boosted models on **100k+ League matches** with **Scikit-learn**, reaching **58% accuracy against a 50% coin-flip baseline** through **feature engineering** and **GridSearchCV**